



Enstatite

△ Piece of massive enstatite suitable for tumble-polishing

While it is an important rock-forming mineral, enstatite's major commercial use is as a gemstone, in colourless, pale yellow, or pale green hues.

The most popular colour is the emerald-green variety, known as chrome-enstatite, and its green colour is caused by traces of chromium – hence the name. All colours are relatively rare, and gemstones are faceted or cut *en cabochon*. Mysore, India, produces star-enstatite; Canada produces iridescent enstatite; and the gem gravels of Myanmar and Sri Lanka yield good-quality facet-grade material.



Enstatite crystal | Rough | Enstatite is rarely found in well-formed crystals such as this, a large specimen that shows the mineral's prismatic form.



Rough specimen | Rough | This piece of enstatite demonstrates its occurrence as a rock-forming mineral. It could be cut and polished, but would not yield a gem.



Polished enstatite pebble | Cut | This tumble-polished baroque of enstatite shows the multicoloured patterning – in this case in a patchwork-like formation – that sometimes occurs when enstatite is found in conjunction with other minerals.

Enstatite in space

Minerals of the early solar system

The presence of enstatite in certain meteorites implies that it was one of the first-formed silicate minerals in the solar nebula that created the Earth and solar system. It occurs in about 10 per cent of meteorites known as chondrites. These are agglomerations of mineral grains that never became part of large enough bodies to undergo melting and re-crystallization. It is thought that they were first formed near the centre of the solar system.



Stony iron meteorite Known as chondrites, this variety of iron meteorite sometimes carries quantities of enstatite.



Cat's-eye cabochon | Cut | Enstatite is one of the minerals that will sometimes produce a star or "cat's eye" when polished *en cabochon*, as in this example.



Oval brilliant | Colour variety | This fine-quality, highly transparent yellow enstatite gemstone has been faceted in a modified brilliant oval cut.